DATA SCIENCE COURSEWORK | PYTHON

Customer Churn & Missing Data

Comparing imputation strategies and their relationship to exploratory churn-model performance.

3,000
customer records

1,420
complete cases

5
imputation / retention strategies

Objective

Evaluate the missingness in customer attributes, compare several ways to retain incomplete records, and examine how the resulting datasets affect Random Forest churn scores.

Approach

The workflow profiles missingness overall and by observed segment, region, and tenure. It compares complete-case deletion, median imputation, KNN, iterative Bayesian-ridge imputation (MICE-style), and iterative ExtraTrees imputation. A 15% artificial mask on complete cases supports a recovery comparison using normalized RMSE, average Kolmogorov-Smirnov distance, and Frobenius distance between correlation matrices. Five-fold stratified cross-validation is then used for the exploratory churn comparison.

Recorded Results

Evaluation	Best recorded method	Value
Point recovery (NRMSE)	MICE-style	0.2660
Marginal shape (average KS)	ExtraTrees	0.0298
Correlation structure (Frobenius)	KNN, k=15	0.1175
Churn ROC-AUC	Complete-case	0.6855

Evaluation	Best recorded method	Value
Churn F1	MICE-style	0.4705

On the saved churn comparison, ROC-AUC spans 0.6777 to 0.6855 across methods. Results vary by evaluation metric, so no single imputation approach dominates.

Interpretation and Limitations

> **Exploratory results only.** The original workflow fits preprocessing and imputers before cross-validation, allowing information to cross fold boundaries. Its churn scores should not be presented as unbiased out-of-sample performance. A production-quality evaluation should fit all preprocessing inside a cross-validation pipeline.

Associations between missingness and observed customer characteristics do not establish MAR. The proposed MNAR explanation for credit-utilization missingness is not identified by the available diagnostics. Artificial masking and the complete-case benchmark also have limitations.

Reproduction

Install dependencies with `python -m pip install -r requirements.txt`, then run `python analysis.py`. The script reads the bundled CSV beside itself and prints tables while displaying diagnostic plots.

Coursework analysis on a supplied dataset. Not a deployed churn model, causal analysis, or validated customer decision system.